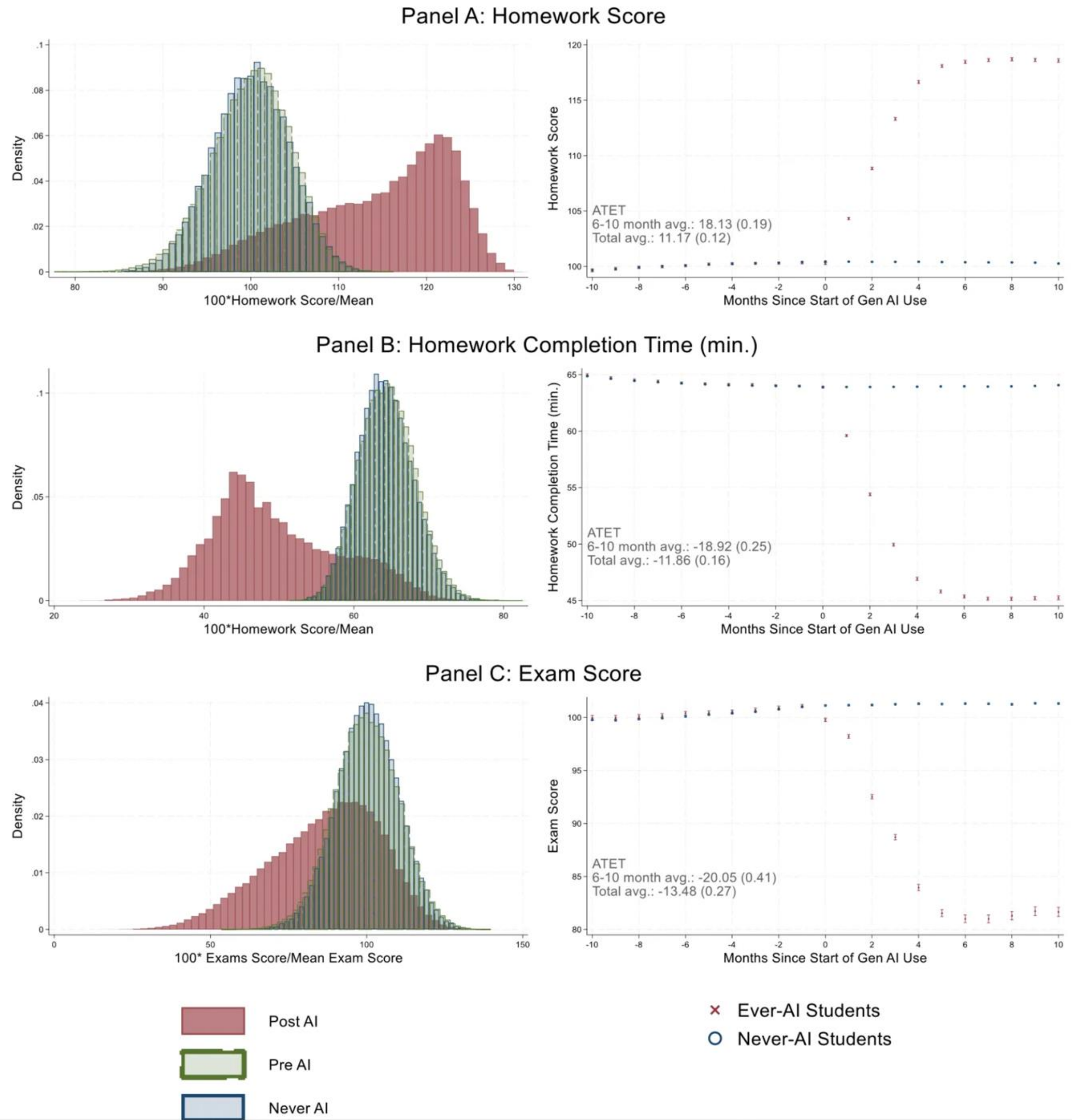
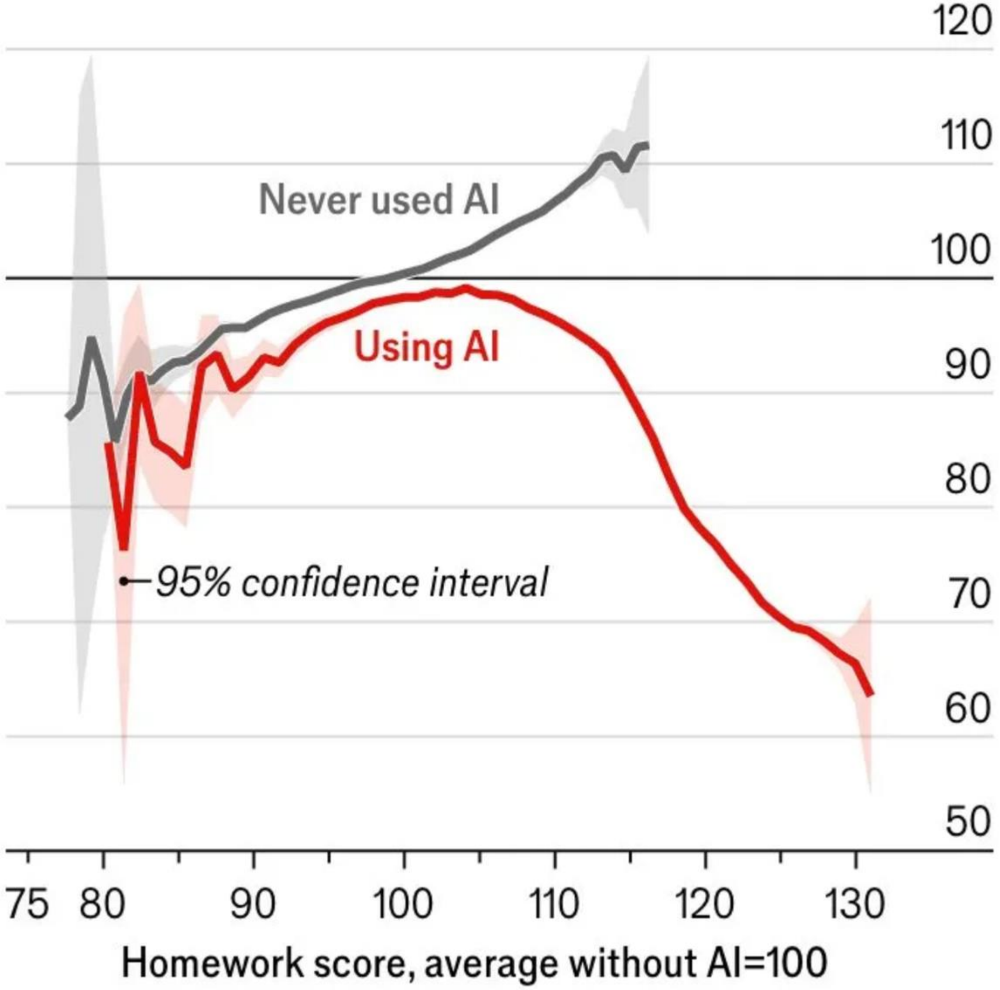
Using AI in ENME 441

Emerging evidence shows AI significantly degrades learning



China*, average exam score

Jan 2023-Jun 2025, average without AI=100

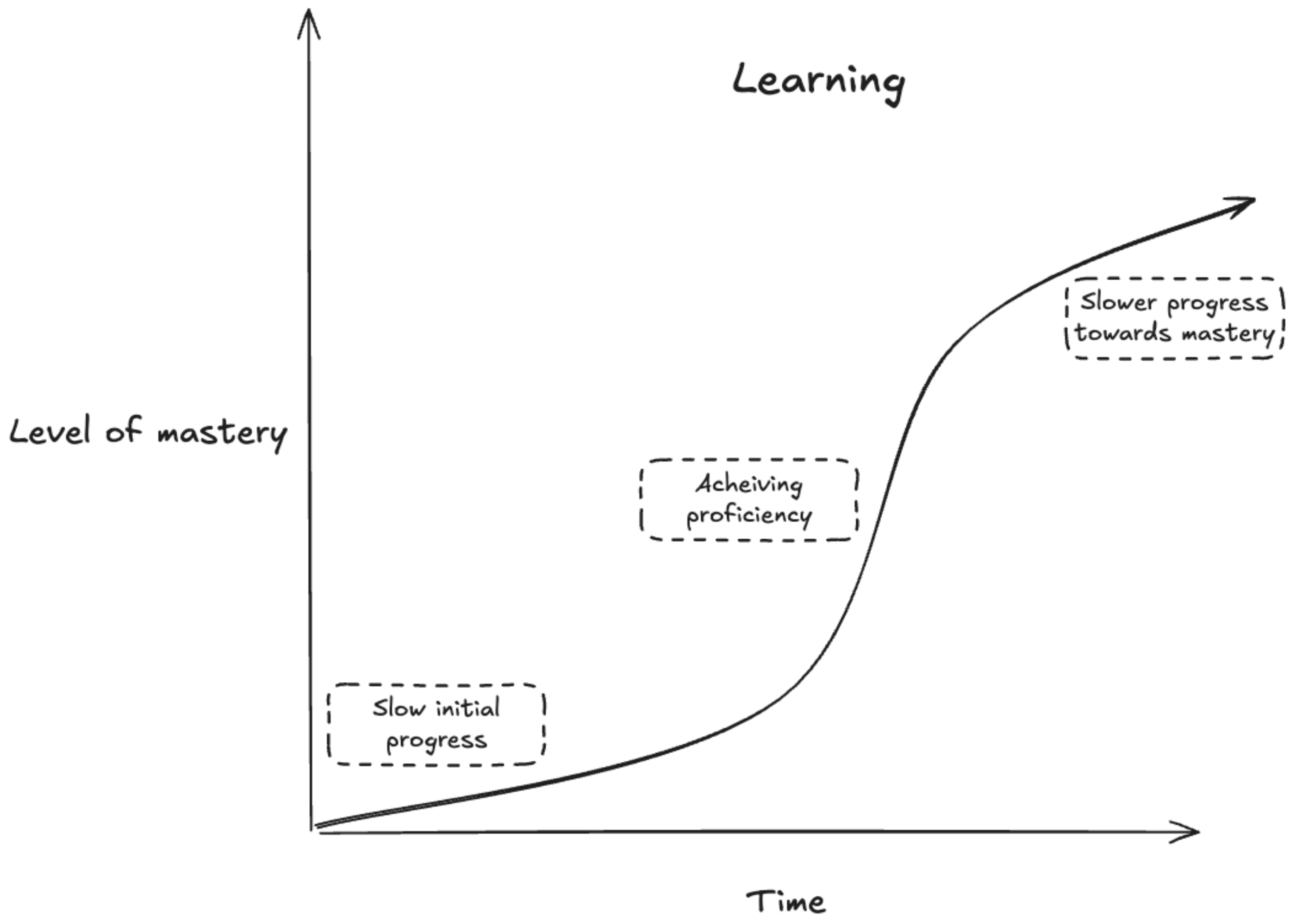


*26,811 pupils aged 12-18

Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

AI is a Floor Raiser, Not a Ceiling Raiser

The learning process



- **Slow initial progress**

  Starting to learn any new topic is always a grind: nothing works, and you can't yet tell why.

- **Achieving proficiency**

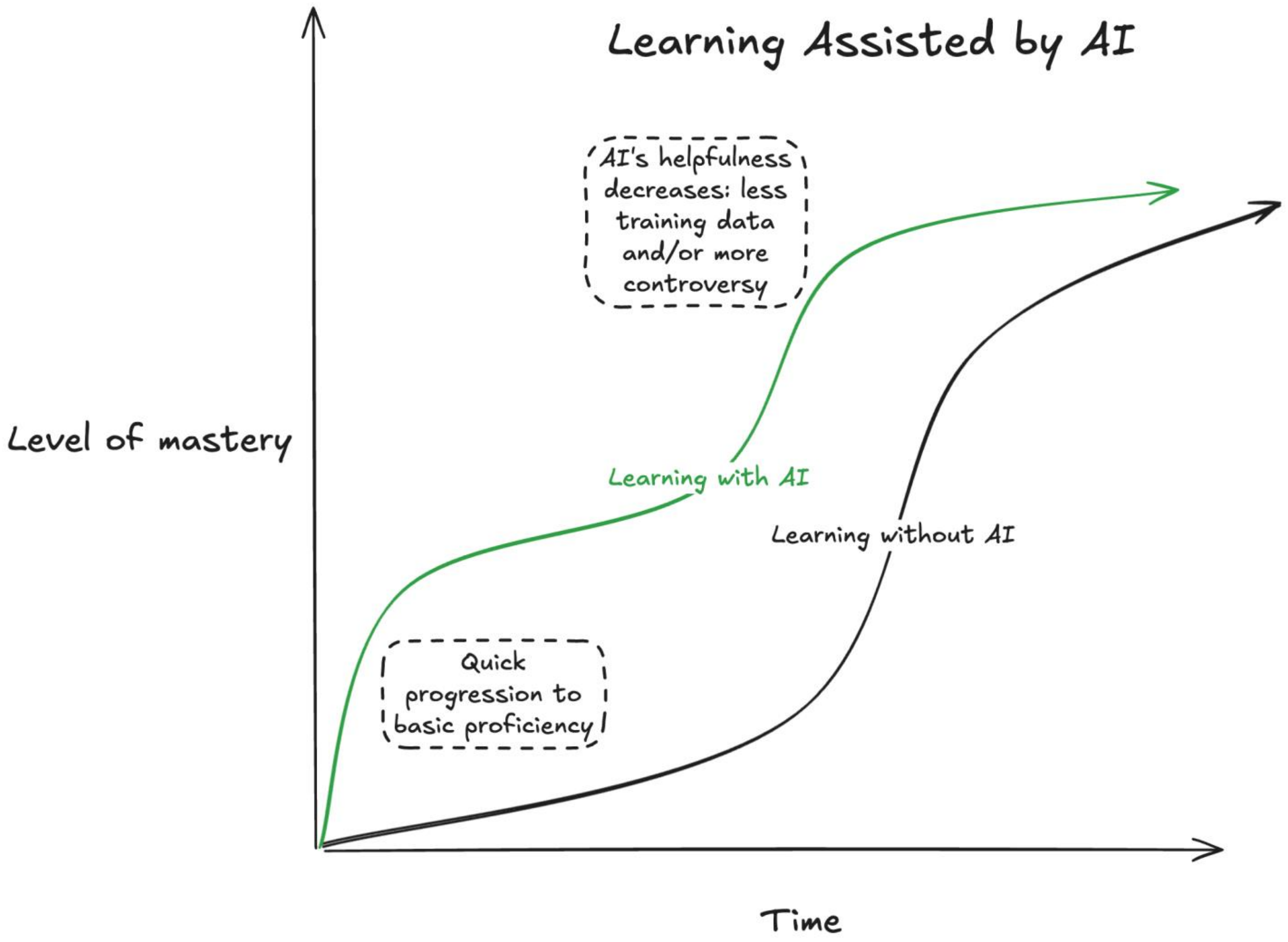
  As the fundamentals take root, the learning curve turns steeply upward.

- **Slow progress toward mastery**

  Gaining mastery is an asymptotic process that takes time (and it is impossible to get to "perfect" mastery of any subject – there is always more to learn)

All figures from:
elroy.bot/blog/2025/07/29/ai-is-a-floor-raiser-not-a-ceiling-raiser.html

Learning assisted by AI



AI enables quick progression to basic proficiency

AI collapses the slow early grind — you get to "it works" far sooner.

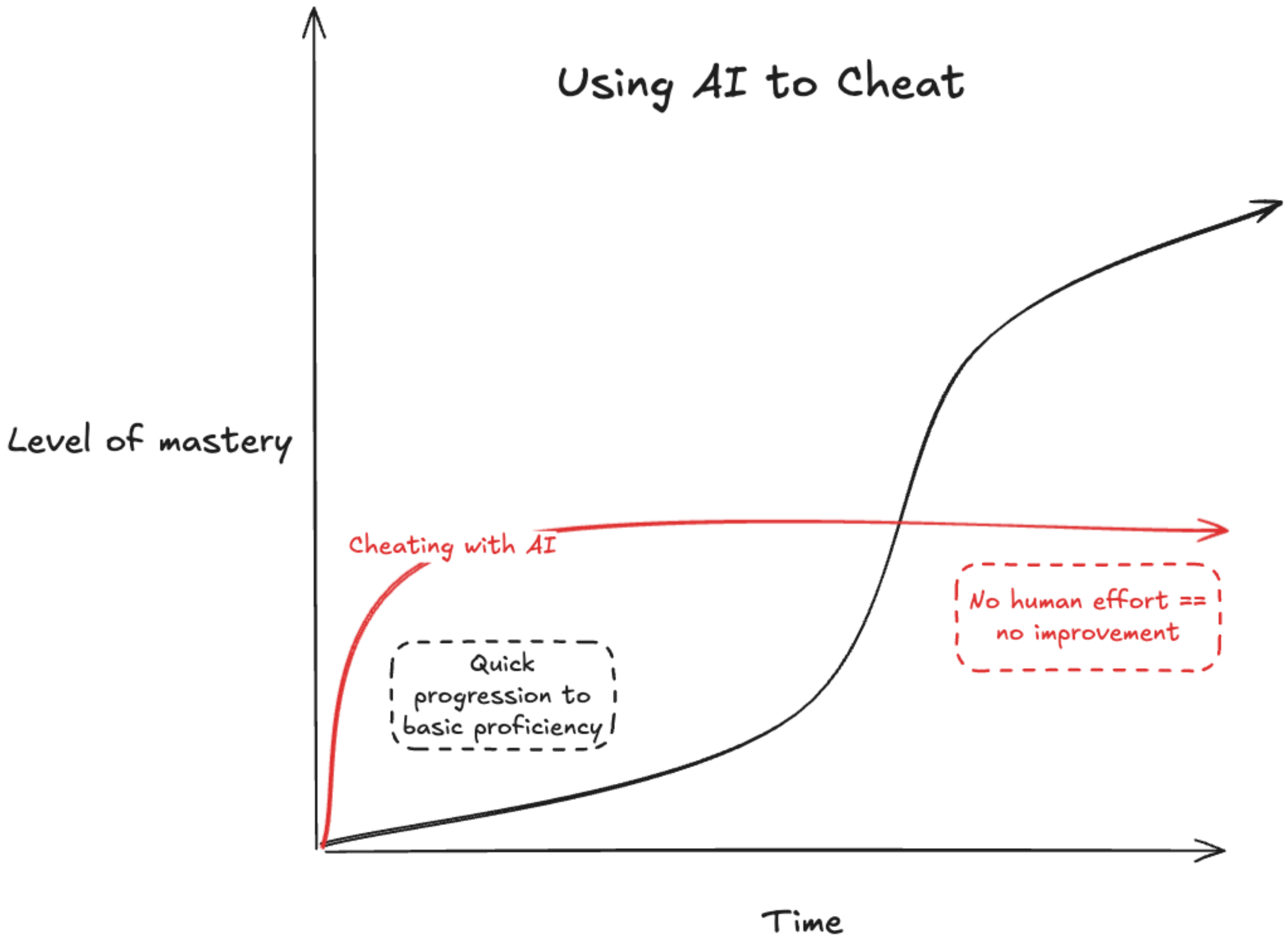
Then its utility decreases

Less training data, more nuance that may be missed by LLMs: the deeper you go, the less it has to say, and the less you can trust the results.

> *"AI is shallow. The deeper I go, the less it seems to be useful."*

THE PLATEAU

Using AI to cheat



Same quick start

Letting AI do all the work yields rapid initial progress.

But…

No human effort == no improvement

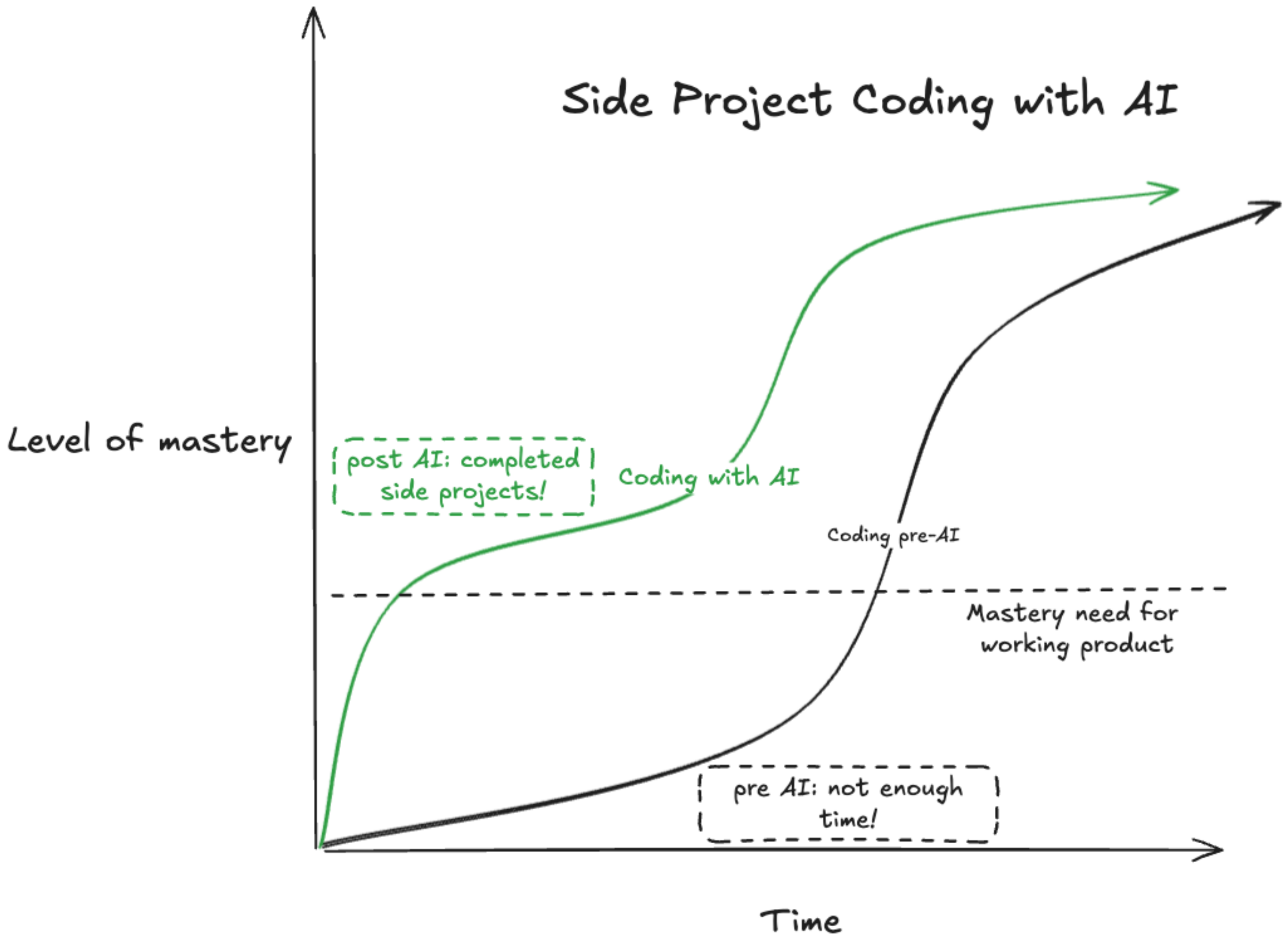
There is no understanding by the human in the loop, because there is no loop!

The curve goes flat exactly where the model's ability stops. You inherit its ceiling as your own (assuming you even make the effort to understand the AI-generated content).

> *"Cheaters will plateau at whatever level the AI can provide — cheaters, in the long run, won't prosper."*

AI RAISES THE FLOOR FOR SMALLER CODING PROJECTS

LLMs are great for small side projects



- **The bar for a working product is low**

  A smaller personal project only has to clear "it works ok and it's useful to me."

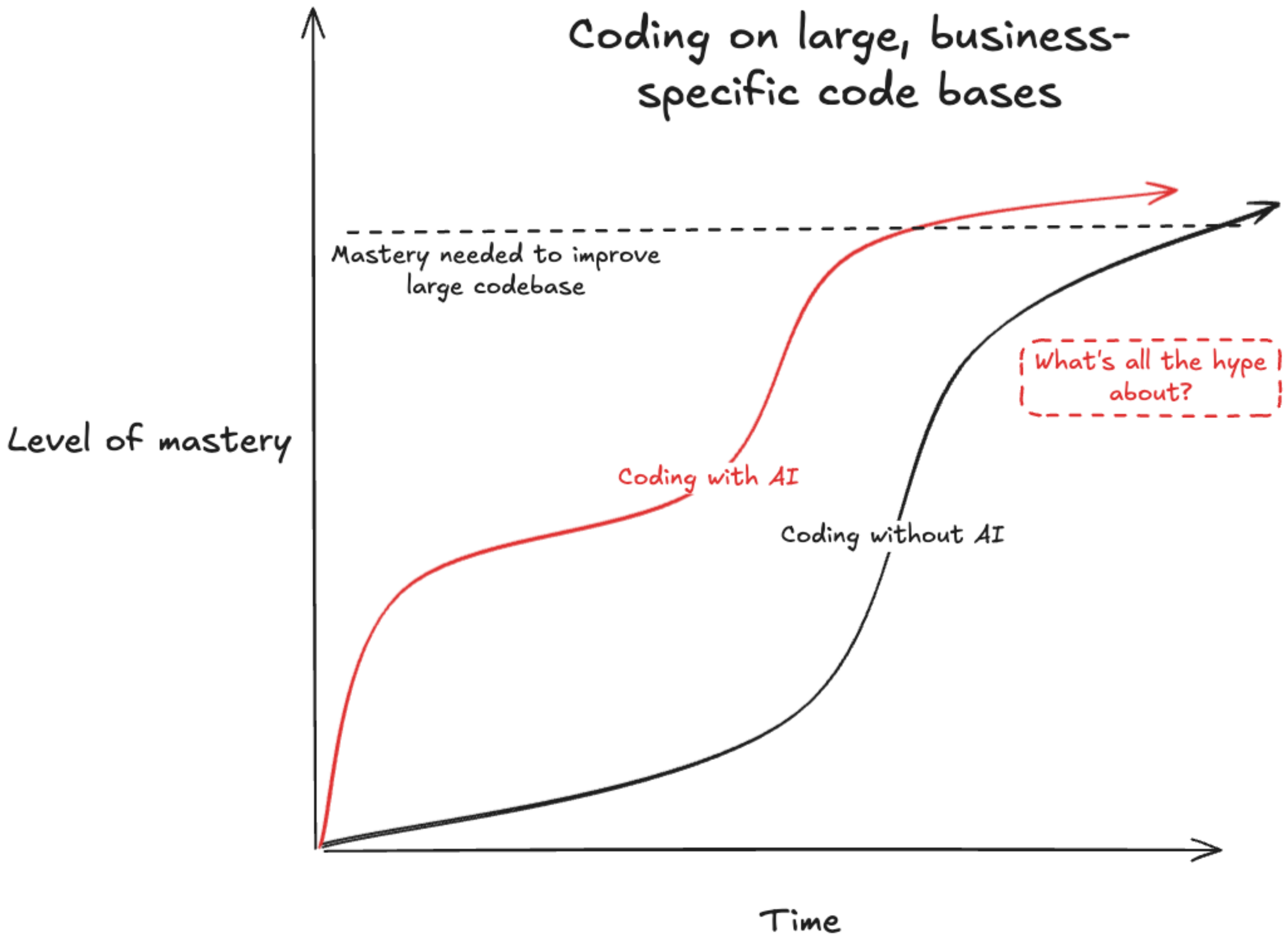
- **Pre-AI: not enough time**

  Projects stall and are left incomplete.

- **Post-AI: small projects get finished – good!**

  The raised floor sits above the threshold — AI enables completion of side projects that would otherwise die.

Large coding projects



The bar sits above AI's reach

Improving large and domain-specific systems demands context the LLM often does not have.

Faster to competency

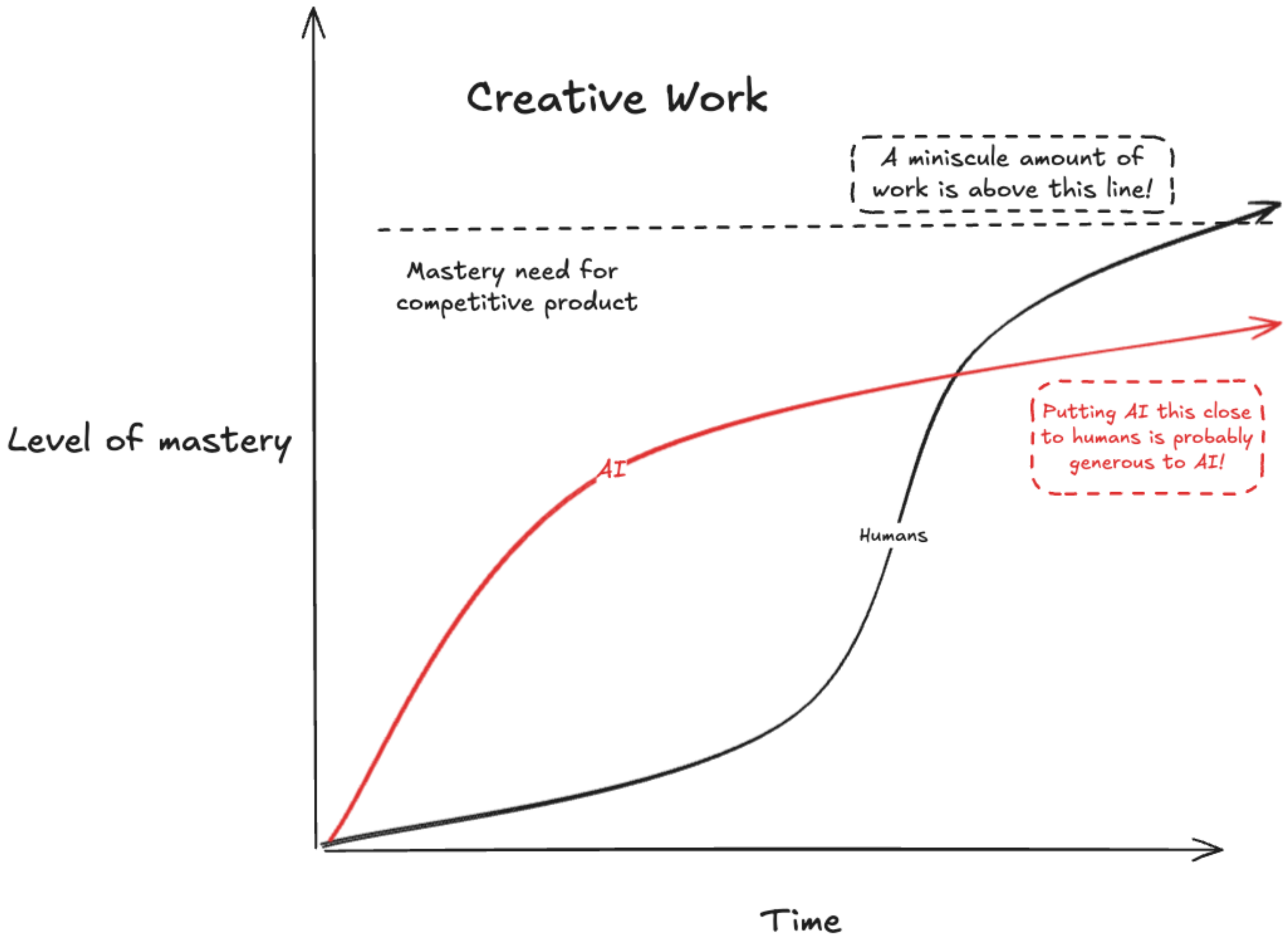
AI moves you up the curve, but return on investment is not infinite, and humans are still needed in the loop.

“What's all the hype about?”

This is the work where AI looks least impressive to experts in the field.

THE CEILING FOR CREATIVE WORK REMAINS ABOVE THE BEST AI MODELS TODAY

Creative work



Competitive work sits above the LLM threshold

Only a minuscule amount of creative output clears the bar that matters.

AI rises fast, then plateaus early

Quick to reach average mastery (“AI slop”), but only incremental improvements at high levels of mastery.

Humans keep climbing

The curve that eventually crosses the line of high mastery is still the human one.

Stack Underflow – Losing the Craft of Coding

> “There’s an argument to be made — a bet, more so — that the hard skills we were once required to master are now and forever obsolete, and thus that the withering away of low-level competencies is unproblematic. **But I would venture that what has always been true in software engineering will remain true: he who understands the layer beneath will thrive.**
>
> Ultimately, engineers debug and build; that’s the job. Of course, a modern-day AI-fluent engineer can point an agent to a problem and say “fix this,” and it might accurately debug the problem, though at other times it will flounder. **It will dive deep down misguided avenues and spit out useless code that tangles the web beyond its own capacity to disentangle it.** This is when the AI-fluent, code-poor engineer hits his ceiling. And then what, nuke the project and start over? This might work at a startup, but not at a tech giant with a codebase comprising billions of lines, where a single deployment failure could cost the company millions of dollars.
>
> […] **college students who delegate their computer science assignments to Claude are doomed**. Coding, quite literally, is a linguistic exercise, and as anyone who “hasn’t spoken Spanish since high school” knows, such fluencies atrophy over time.

https://www.thenewatlantis.com/publications/stack-underflow-losing-the-craft-of-coding
Jarett Malouf
Aug 2026

THE TAKEAWAY

Use AI to raise the floor, but the ceiling is yours to master.

Use AI to get unstuck

Boilerplate, syntax, a first version that runs, dealing with error messages, simplifying code – handing these steps to AI raises the floor, and is where LLMs are reliably excellent.

Cheating sets your ceiling

Hand an assignment to a model and you plateau at whatever the model can provide, even if you scrutinize the model output in detail.

Depth is still the human domain

The deeper the problem, the thinner the training data. Gaining mastery over development of your own hardware and code (or any other intellectual domain outside of programming) cannot be given to you by LLMs.

Source: `elroy.bot/blog/2025/07/29/ai-is-a-floor-raiser-not-a-ceiling-raiser.html`